

Active ants

$$\begin{cases} \mathrm{d}X_t^i &= \mathrm{Pev}(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i &= \chi F\left(\left(X_t^i, \theta_t^i\right), \mu_t^N\right)\mathrm{d}t + \sqrt{2}\mathrm{d}B_t^i \end{cases}$$

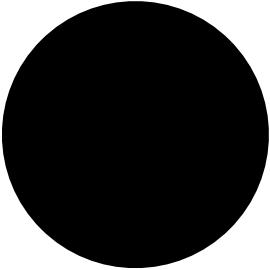
- Ant model (dW, Bruna & Burger, SIAM J. Appl. Dyn. Syst., 2025):

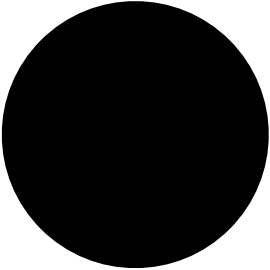
$$F = n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j)$$

- ∇K attractive interaction kernel

$$\bullet \quad n(\theta) = (-\sin(\theta), \cos(\theta))^{\top} = v(\theta)^{\perp}$$

$$\bullet \text{ In 2D: } v^\perp \cdot (\cdot) \equiv \text{proj}_{v^\perp}(\cdot)$$



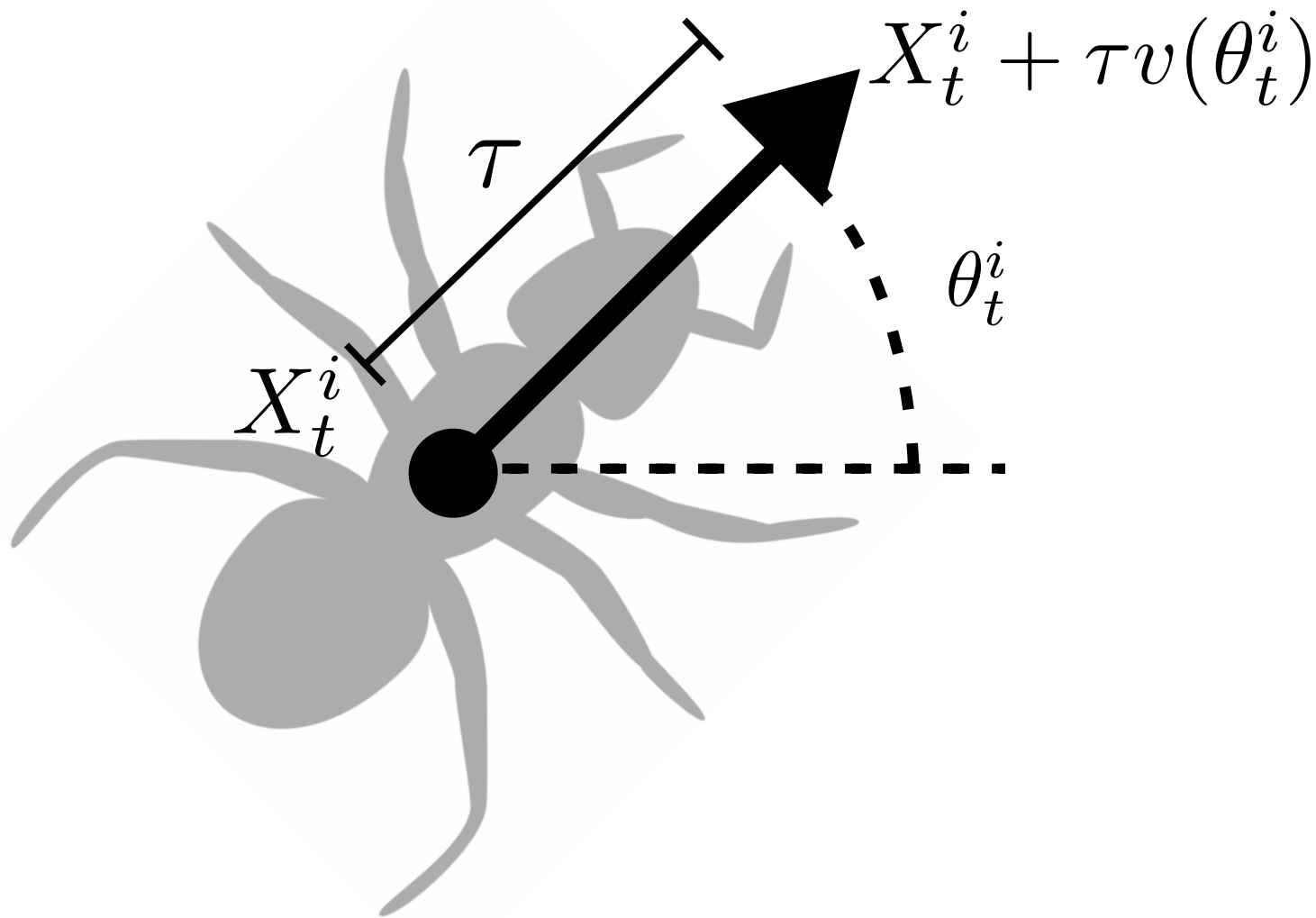








$$\nabla K(x) - v$$



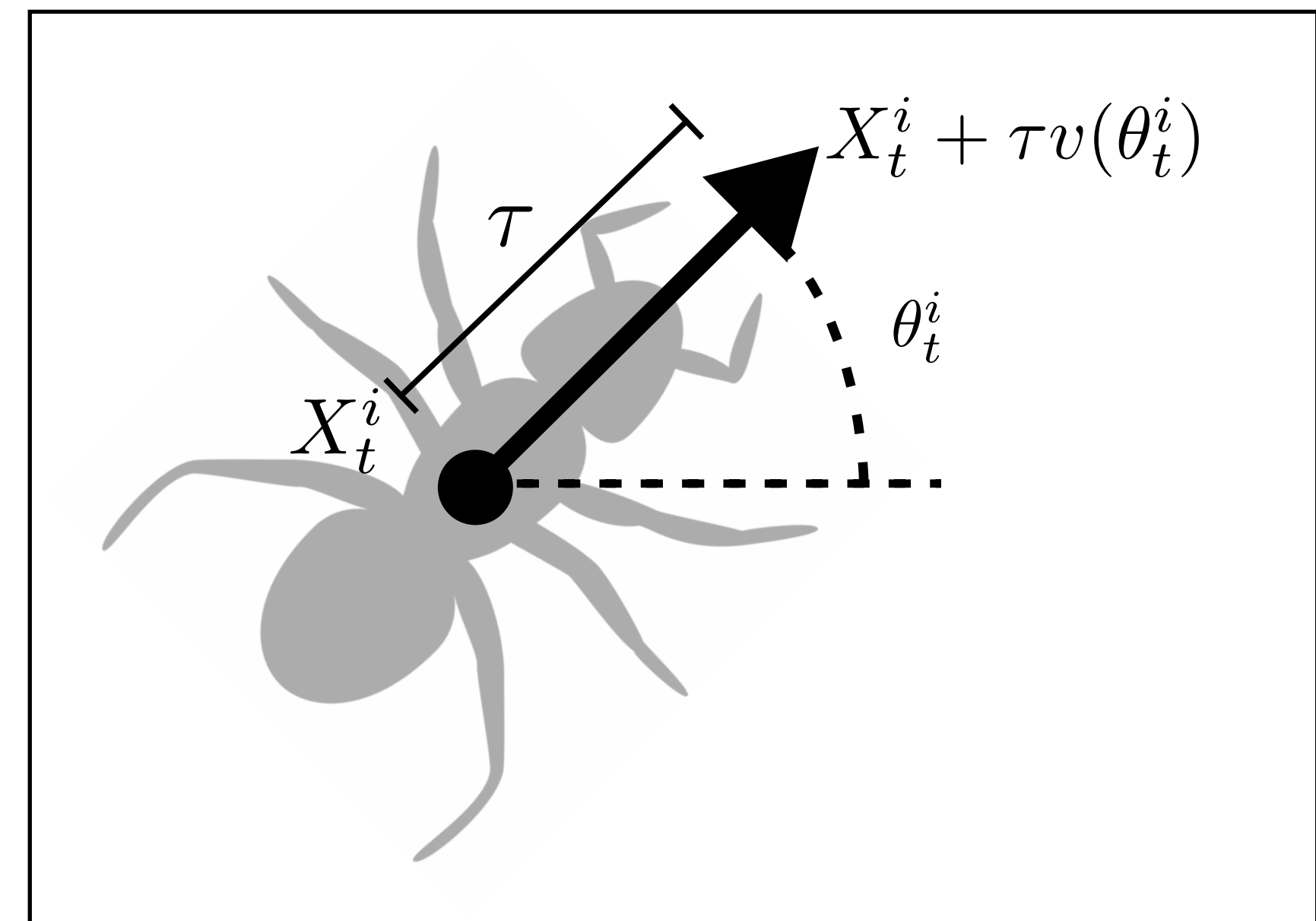
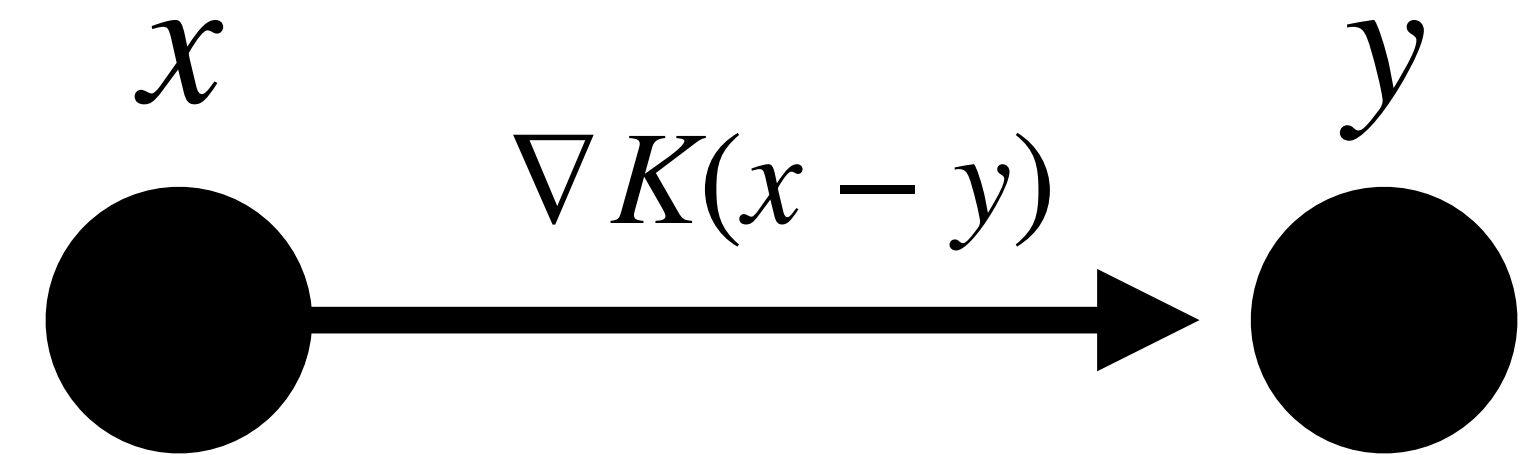
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- Ant model (dW, Bruna & Burger, SIAM J. Appl. Dyn. Syst., 2025):

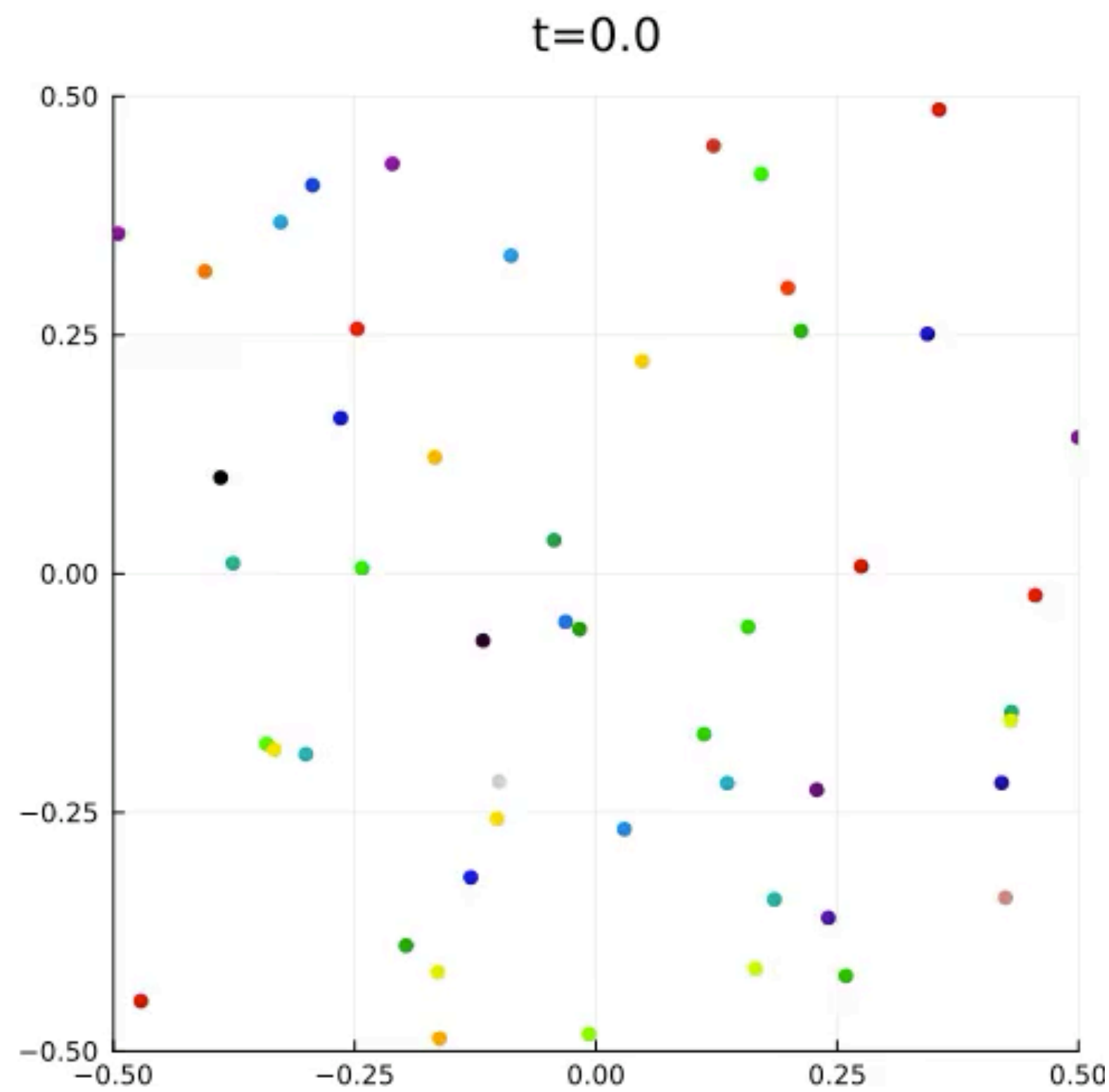
$$F = n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j)$$

- ∇K attractive interaction kernel
- $n(\theta) = (-\sin(\theta), \cos(\theta))^T = v(\theta)^\perp$
- In 2D: $v^\perp \cdot (\cdot) = \text{proj}_{v^\perp}(\cdot)$

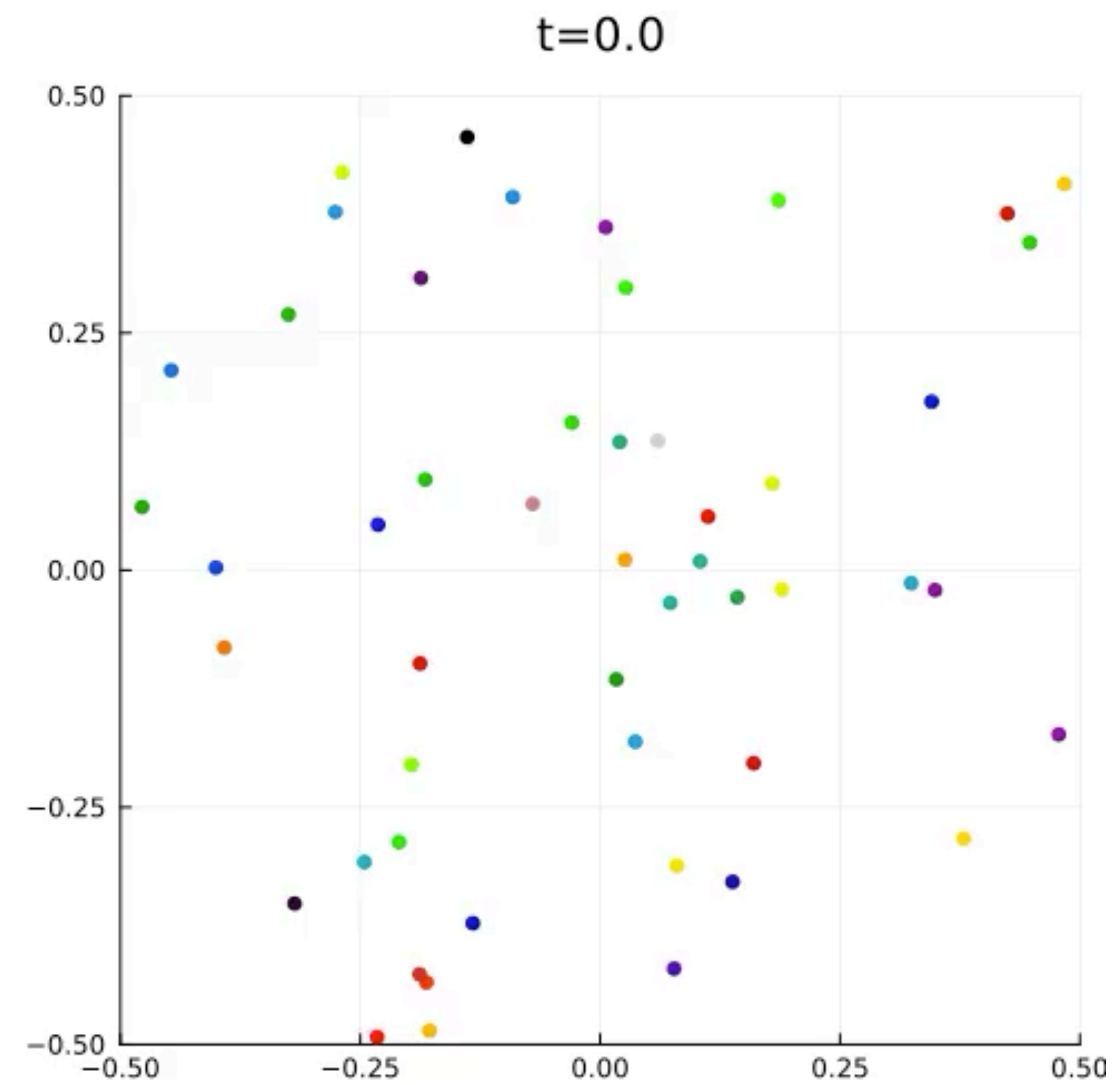


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- $F = n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j)$ on $\mathbb{T}^2$



$\tau = 0$



$\tau = 0.15$